

AWS COVID-19 End-to-End Data Pipeline Project

PROJECT OVERVIEW

This project builds a complete AWS end-to-end data engineering pipeline for COVID-19 datasets. The project includes two tasks:

Task 1 – Data Warehousing & Analytics Enablement

- Build Gold-layer dimensional models
- Load curated data into Amazon Redshift
- Enable analytical reporting using Athena
- Run SQL-based BI reporting queries

Task 2 – Data Ingestion & Transformation

- Upload raw COVID datasets into Amazon S3
- Clean and standardize data using AWS Glue PySpark
- Create partitioned Silver-layer parquet datasets

STEP 1 – OPEN AWS CONSOLE

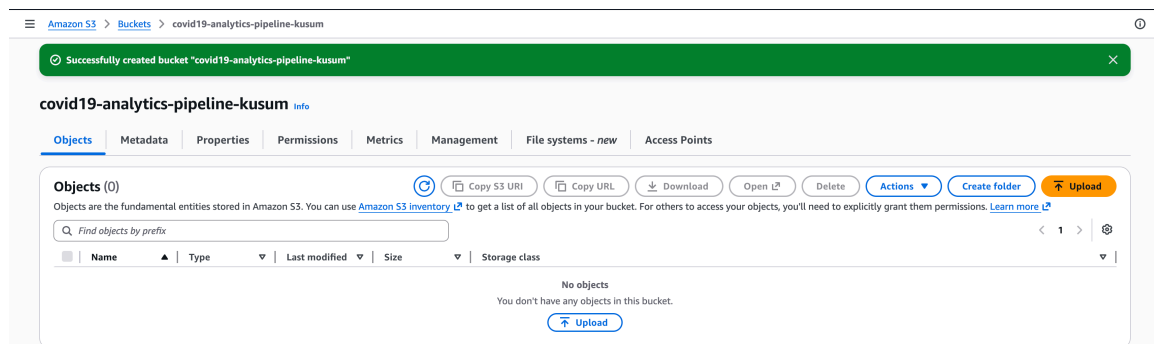
- Sign in to your AWS account.

STEP 2 – CREATE S3 BUCKET

- I opened Amazon S3 from the AWS Console Search and Create bucket.

Bucket name:

“covid19-analytics-pipeline-kusum”



STEP 3 – CREATE S3 FOLDER STRUCTURE:

Inside the bucket, I created the following folders:

raw/

silver/
gold/
rejected/
scripts/
athena-results/

Amazon S3 > Buckets > covid19-analytics-pipeline-kusum

Successfully created folder "athena-results".

covid19-analytics-pipeline-kusum

Objects | Metadata | Properties | Permissions | Metrics | Management | File systems - new | Access Points

Objects (6)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
athena-results/	Folder	-	-	-
gold/	Folder	-	-	-
raw/	Folder	-	-	-
rejected/	Folder	-	-	-
scripts/	Folder	-	-	-
silver/	Folder	-	-	-

Inside raw/ create:

- covid_cases/
- covid_vaccinations/
- covid_testing/

Amazon S3 > Buckets > covid19-analytics-pipeline-kusum > raw/

Successfully created folder "covid_testing".

raw/

Objects | Properties

Objects (3)

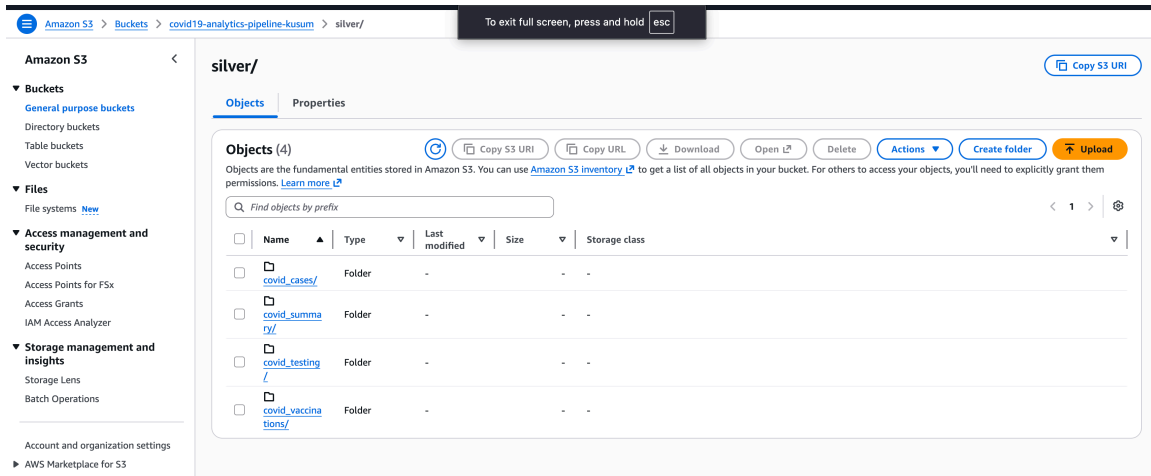
Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
covid_cases/	Folder	-	-	-
covid_testing/	Folder	-	-	-
covid_vaccination s/	Folder	-	-	-

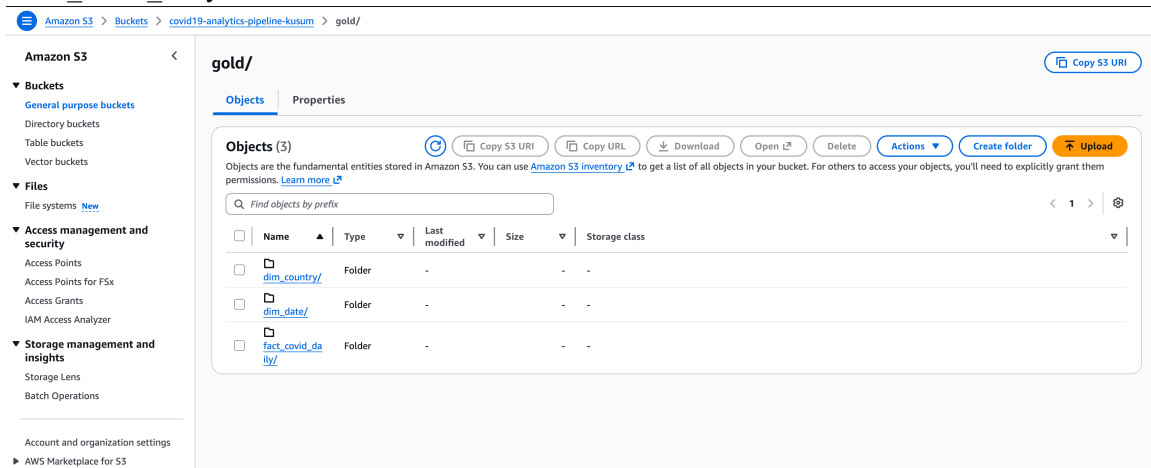
Inside silver/ create:

- covid_cases/
- covid_vaccinations/
- covid_testing/
- covid_summary/



Inside gold/ create:

- dim_country/
- dim_date/
- fact_covid_daily/



STEP 4 – CREATE CSV FILES

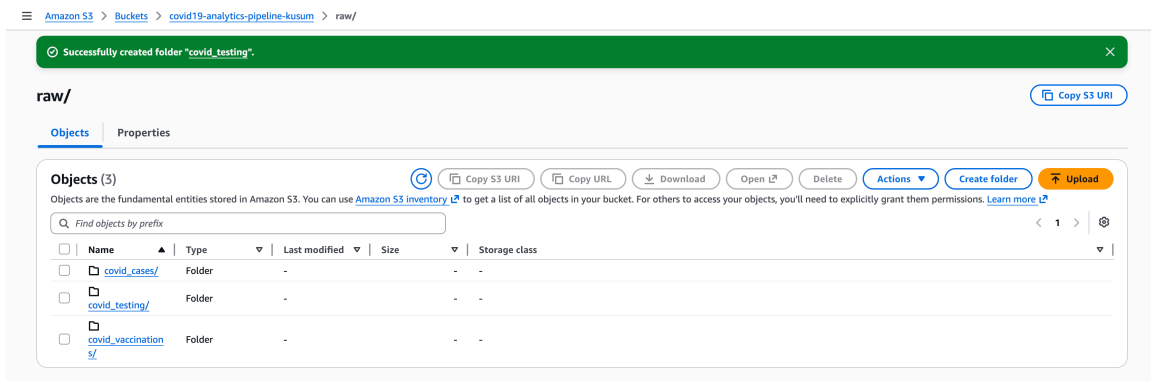
- Create 3 CSV files using Notepad or VS Code.

1. covid_cases.csv
2. covid_vaccinations.csv
3. covid_testing.csv

Upload into:

raw/covid_cases/
raw/covid_vaccinations/
raw/covid_testing/

Take Screenshot:



STEP 5 – CREATE IAM ROLE

creating the IAM ROLES;

Search:

IAM

Go to:

Roles → Create role

Choose:

AWS Service → Glue

Attach policies:

- AmazonS3FullAccess
- AWSGlueConsoleFullAccess
- AmazonAthenaFullAccess
- AmazonRedshiftFullAccess

Role name:

“GlueCovidDataPlatformRole”

This role allowed AWS Glue to access S3, Glue Catalog, Athena, and Redshift-related services.

Take Screenshot:

Identity and Access Management (IAM)

Dashboard

▼ Access Management

- Roles
- Policies
- IAM users
- IAM user groups
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

▼ Access reports

- Access Analyzer
- Resource analysis
- Unused access
- Analyzer settings
- Credential report
- Organization activity
- Service control policies
- Resource control policies

Role GlueCovidDataPlatformRole created.

Roles (1/6) [Info](#)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Q Search IAM

Role name	Trusted entities	Last activity
AWSServiceRoleForResourceExplorer	AWS Service: resource-explorer-2 (Service-Linker)	31 minutes ago
AWSServiceRoleForSupport	AWS Service: support (Service-Linker)	-
AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service-Linker)	-
☒ GlueCovidDataPlatformRole	AWS Service: glue	-
myLambda	AWS Service: lambda	-
s3-roles	AWS Service: ec2	20 days ago

Roles Anywhere [Info](#) [Manage](#)

Authenticate your non AWS workloads and securely provide access to AWS services.

Access AWS from your non AWS workloads

Operate your non AWS workloads using the same authentication and authorization strategy that you use within AWS.

X.509 Standard

Use your own existing PKI infrastructure or use [AWS Certificate Manager Private Certificate Authority](#) to authenticate identities.

Temporary credentials

Use temporary credentials with ease and benefit from the enhanced security they provide.

Identity and Access Management (IAM)

Dashboard

▼ Access Management

- Roles
- Policies
- IAM users
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▼ Access reports

- Access Analyzer
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- Service control policies
- Resource control policies

GlueCovidDataPlatformRole [Info](#) [Delete](#)

Allows Glue to call AWS services on your behalf.

Summary [Edit](#)

Creation date
June 01, 2026, 20:58 (UTC-05:00)

ARN
[arn:aws:iam::262892773233:role/GlueCovidDataPlatformRole](#)

Last activity
-

Maximum session duration
1 hour

Permissions | Trust relationships | Tags | Last Accessed | Revoke sessions

Permissions policies (4) [Info](#) [Simulate](#) [Remove](#) [Add permissions](#)

You can attach up to 10 managed policies.

Q Search IAM

Filter by Type
All types

Policy name	Type	Attached entities
AmazonAthenaFullAccess	AWS managed	1
AmazonRedshiftFullAccess	AWS managed	1
AmazonS3FullAccess	AWS managed	1
AWSGlueConsoleFullAccess	AWS managed	1

STEP 6 – CREATE GLUE DATABASE

- Open AWS Glue.

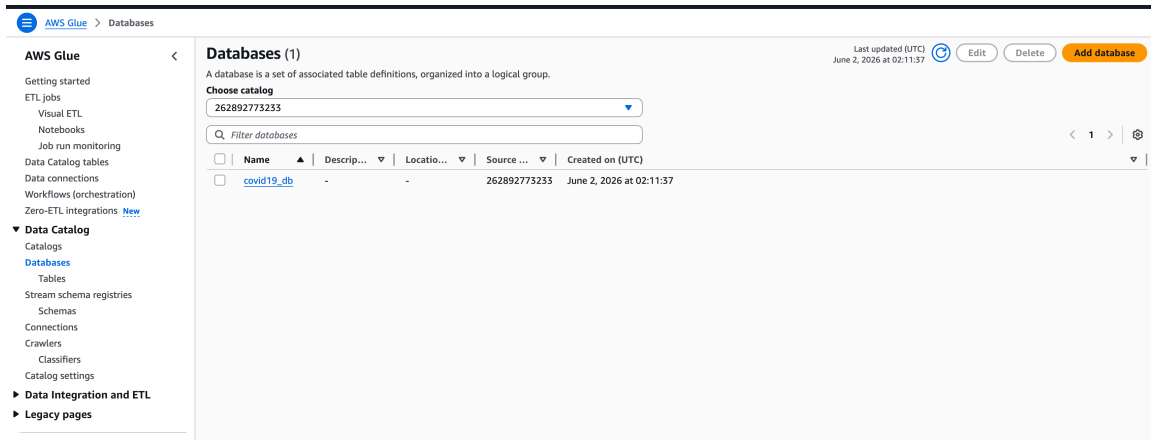
Go to:

Data Catalog → Databases → Add database

Database name:

“covid19_db”

Take Screenshot:



STEP 7 – CREATE RAW CRAWLER

Go to:

Glue → Crawlers → Create crawler

Crawler name:

covid19_raw_crawler

Add paths:

s3://covid19-analytics-pipeline-kusum/raw/covid_cases/

s3://covid19-analytics-pipeline-kusum/raw/covid_vaccinations/

s3://covid19-analytics-pipeline-kusum/raw/covid_testing/

Select role:

“GlueCovidDataPlatformRole”

Target database:

‘covid19_db’

Run crawler.

Expected tables:

- covid_cases
- covid_vaccinations
- covid_testing

Screenshots:

AWS Glue

Crawlers

>

Add crawler

AWS Glue

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Zero-ETL integrations

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Step 1

Set crawler properties

Step 2

Choose data sources and classifiers

Step 3

Configure security settings

Step 4

Set output and scheduling

Step 5

Review and create

Review and create

Step 1: Set crawler properties

Edit

Set crawler properties

Name

covid19_gold_crawler

Description

-

Tags

-

Step 2: Choose data sources and classifiers

Edit

Data sources (3)

The list of data sources to be scanned by the crawler.

Type

Data source

Parameters

S3

s3://covid19-analy...

Recrawl all

S3

s3://covid19-analy...

Recrawl all

S3

s3://covid19-analy...

Recrawl all

Step 3: Configure security settings

Edit

Configure security settings

IAM role

GlueCovidDataPlatformRole

Security configuration

-

Lake Formation configuration

-

Step 4: Set output and scheduling

Edit

Set output and scheduling

Database

covid19_db

Table prefix - optional

-

Maximum table threshold - optional

-

Schedule

On demand

Cancel

Previous

Create crawler

AWS Glue

Crawlers

>

covid19_raw_crawler

AWS Glue

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Crawler successfully starting

The following crawler is now starting: "covid19_raw_crawler"

0

0

0

2

0

0

0

0

0

0

covid19_raw_crawler

Last updated (UTC)

June 2, 2026 at 02:15:14

Run crawler

Edit

Delete

Crawler properties

Name

covid19_raw_crawler

Description

-

Maximum table threshold

-

IAM role

GlueCovidDataPlatformRole

Security configuration

-

Database

covid19_db

Lake Formation configuration

-

State

READY

Table prefix

-

Advanced settings

Crawler runs

Schedule

Data sources

Classifiers

Tags

Crawler runs (1/1)

The list of crawler runs for this crawler.

Stop run

View CloudWatch logs

View run details

Filter dots

Filter by a date and time range

Start time (UTC)

End time (UTC)

Current/last duration

Status

DPU hours

Table changes

June 2, 2026 at 02:15:40

-

-

Running

-

-

AWS Glue > **Crawlers** > covid19_raw_crawler

AWS Glue

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covid19_raw_crawler Last updated (UTC) June 2, 2026 at 02:18:57 [Run crawler](#) [Edit](#) [Delete](#)

Crawler successfully starting
The following crawler is now starting: "covid19_raw_crawler"

Crawler properties

Name covid19_raw_crawler	IAM role GlueCovidDataPlatformRole	Database covid19_db	State READY
Description -	Security configuration -	Lake Formation configuration -	Table prefix -
Maximum table threshold -			

► **Advanced settings**

Crawler runs | Schedule | Data sources | Classifiers | Tags

Crawler runs (1) Stop run View CloudWatch logs View run details

The list of crawler runs for this crawler.

	Start time (UTC)	End time (UTC)	Current/last duration	Status	DPU hours	Table changes
☐	June 2, 2026 at 02:15:40	June 2, 2026 at 02:16:23	43 s	Completed	0.124	3 table changes, 0 partition changes

AWS Glue > **Tables**

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Announcing new optimization features for Apache Iceberg tables
Optimize storage for Apache Iceberg tables with automatic snapshot retention and orphan file deletion. [Learn more](#)

Tables
A table is the metadata definition that represents your data, including its schema. A table can be used as a source or target in a job definition.

Tables (3) Last updated (UTC) June 2, 2026 at 02:19:30 [Delete](#) [Add tables using crawler](#) [Add table](#)

View and manage all available tables.

Choose catalog
262892773233

Choose database
covid19_db

☐	Name	Database	Location	Classification	Deprecated	View data	Data quality	Column statistics
☐	covid_cases	covid19_db	s3://covid19-analytics-	UNKNOWN	-	Table data	View data quality	View statistics
☐	covid_testing	covid19_db	s3://covid19-analytics-	UNKNOWN	-	Table data	View data quality	View statistics
☐	covid_vaccinations	covid19_db	s3://covid19-analytics-	UNKNOWN	-	Table data	View data quality	View statistics

STEP 8 – CREATE GLUE ETL JOB

Go to:

Glue → ETL Jobs → Script editor

Choose:

- Spark
- Python

Job name:

“covid19_silver_gold_etl_job”

Select IAM role:

GlueCovidDataPlatformRole

Paste PySpark ETL code.

The ETL job performs:

- Duplicate removal
- Null handling
- Data cleansing
- Date conversion
- Partition column creation
- Silver-layer creation
- Gold-layer dimensional model creation

Run the job.

Wait until:

Succeeded

Screenshots:

The screenshot displays the AWS Glue console interface. At the top, a green notification bar states "Successfully started job" for the job named "covid19_silver_gold_etl_job". Below this, the job details are shown, including the job name and a timestamp indicating it was last modified on 6/1/2026 at 9:22:12 PM. The "Script" tab is selected, showing the PySpark code used for the ETL process. The code includes imports for sys, aws glue context, job, utils, pyspark context, and sql functions. It defines arguments, initializes the SparkContext, sets up the GlueContext, and defines paths for raw, silver, and gold data layers. The status bar at the bottom indicates the job is in a "Succeeded" state with 0 errors and 0 warnings.

```
1 import sys
2 from aws glue.context import GlueContext
3 from aws glue.job import Job
4 from aws glue.utils import getResolvedOptions
5 from pyspark.context import SparkContext
6 from pyspark.sql.functions import col, to_date, year, month, when, monotonically_increasing_id
7
8
9 args = getResolvedOptions(sys.argv, ["JOB_NAME"])
10
11 sc = SparkContext()
12 glueContext = GlueContext(sc)
13 spark = glueContext.spark_session
14
15 job = Job(glueContext)
16 job.init(args["JOB_NAME"], args)
17
18
19 bucket_name = "covid19-data-platform-kusum"
20
21
22 cases_path = f"s3://{bucket_name}/raw/covid_cases/"
23 vaccinations_path = f"s3://{bucket_name}/raw/covid_vaccinations/"
24 testing_path = f"s3://{bucket_name}/raw/covid_testing/"
25
26 silver_cases_path = f"s3://{bucket_name}/silver/covid_cases/"
27 silver_vaccinations_path = f"s3://{bucket_name}/silver/covid_vaccinations/"
28 silver_testing_path = f"s3://{bucket_name}/silver/covid_testing/"
29 silver_summary_path = f"s3://{bucket_name}/silver/covid_summary/"
30
31 gold_dim_country_path = f"s3://{bucket_name}/gold/dim_country/"
```

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covid19_silver_gold_etl_job Last modified on 6/1/2026, 9:50:24 PM [Actions](#) [Save](#) [Run](#)

Script Job details **Runs** Data quality Schedules Version Control

Job runs (1/3) [View details](#) [Stop job run](#) [Troubleshoot with AI](#) [Table View](#) [Card View](#)

Filter job runs by property

Run status	Retries	Start time (Local)	End time (Local)	Duration	Capacit...	Worker type	Glue version
☒ Running	0	06/01/2026 21:54:33	-	48 s	10 DPUs	G.1X	5.1
☐ Succeeded	0	06/01/2026 21:50:06	06/01/2026 21:51:57	1 m 44 s	10 DPUs	G.1X	5.1
☐ Failed	0	06/01/2026 21:22:38	06/01/2026 21:23:47	56 s	10 DPUs	G.1X	5.1

Run details Input arguments (11) Logs Run insights Metrics Troubleshooting analysis Spark UI [Stop job run](#)

Job name: covid19_silver_gold_etl_job
Id: jr_c52381cf7be4c3c8b7972ddf650a57d8fdd001eca - a4519d4339ef997145124db [f](#)
Run status: ☒ **Running**
Retry attempt number: 0
Initial run: 48 seconds
Trigger name: -
Job run queuing: False

Start time (Local): 06/01/2026 21:54:33
End time (Local): -
Start-up time: -
Execution time: -
Security configuration: -

Glue version: 5.1
Worker type: G.1X
Max capacity: 10 DPUs
Execution class: Standard
Cloudwatch logs: [Output logs](#) [Error logs](#)

Last modified on (Local): 06/01/2026 21:54:36
Log group name: /aws-glue/jobs
Number of workers: 10
Timeout: 480 minutes
Usage profile: -

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covid19_silver_gold_etl_job Last modified on 6/1/2026, 9:50:24 PM [Actions](#) [Save](#) [Run](#)

Script Job details **Runs** Data quality Schedules Version Control

Job runs (1/3) [View details](#) [Stop job run](#) [Troubleshoot with AI](#) [Table View](#) [Card View](#)

Filter job runs by property

Run status	Retries	Start time (Local)	End time (Local)	Duration	Capacit...	Worker type	Glue version
☒ Succeeded	0	06/01/2026 21:54:33	06/01/2026 21:56:27	1 m 48 s	10 DPUs	G.1X	5.1
☐ Succeeded	0	06/01/2026 21:50:06	06/01/2026 21:51:57	1 m 44 s	10 DPUs	G.1X	5.1
☐ Failed	0	06/01/2026 21:22:38	06/01/2026 21:23:47	56 s	10 DPUs	G.1X	5.1

Run details Input arguments (11) Logs Run insights Metrics Troubleshooting analysis Spark UI

Job name: covid19_silver_gold_etl_job
Id: jr_c52381cf7be4c3c8b7972ddf650a57d8fdd001eca - a4519d4339ef997145124db [f](#)
Run status: ☒ **Succeeded**
Retry attempt number: 0
Initial run: 1 minute 48 seconds
Trigger name: -
Job run queuing: False

Start time (Local): 06/01/2026 21:54:33
End time (Local): 06/01/2026 21:56:27
Start-up time: 6 seconds
Execution time: 1 minute 48 seconds
Security configuration: -

DPU Hours: 0.3
Glue version: 5.1
Worker type: G.1X
Max capacity: 10 DPUs
Execution class: Standard
Cloudwatch logs: [Output logs](#) [Error logs](#)

Last modified on (Local): 06/01/2026 21:56:27
Log group name: /aws-glue/jobs
Number of workers: 10
Timeout: 480 minutes
Usage profile: -

STEP 9 – VERIFY SILVER OUTPUT

After the Glue job succeeded, I opened the S3 bucket and checked the Silver-layer folders. Check folders:

- silver/covid_cases/
- silver/covid_vaccinations/
- silver/covid_testing/
- silver/covid_summary/

Expected output:

year=2024/

month=1/

part-0000.snappy.parquet

Screenshots:

Amazon S3 > Buckets > covid19-analytics-pipeline-kusum > silver/ > covid_cases/

Amazon S3

- ▼ Buckets
 - General purpose buckets
 - Directory buckets
 - Table buckets
 - Vector buckets
- ▼ Files
 - File systems [New](#)
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- ▼ Storage management and insights
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 - Batch Operations

Account and organization settings
▶ AWS Marketplace for S3

covid_cases/ [Copy S3 URI](#)

Objects (1) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open L2](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	year=2024/	Folder	-	-	-

Amazon S3 > Buckets > covid19-analytics-pipeline-kusum > silver/ > covid_cases/ > year=2024/ [Copy S3 URI](#)

Amazon S3

- ▼ Buckets
 - General purpose buckets
 - Directory buckets
 - Table buckets
 - Vector buckets
- ▼ Files
 - File systems [New](#)
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Account and organization settings
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year=2024/ [Copy S3 URI](#)

Objects (1) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open L2](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	month=1/	Folder	-	-	-

Amazon S3 > Buckets > covid19-analytics-pipeline-kusum > silver/ > covid_cases/ > year=2024/ > month=1/

Amazon S3

- ▼ Buckets
 - General purpose buckets
 - Directory buckets
 - Table buckets
 - Vector buckets
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month=1/

Objects (1)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	part-0000-5f6a2458-1dd8-4176-b673-c06b2d702b95.c000.snap.py.parquet	parquet	June 1, 2026, 21:55:43 (UTC-05:00)	2.1 KB	Standard

STEP 10 – VERIFY GOLD OUTPUT

Check folders:

gold/dim_country/
gold/dim_date/
gold/fact_covid_daily/

These are Gold-layer dimensional models.

Screenshots:

Amazon S3 > Buckets > covid19-analytics-pipeline-kusum > gold/ > dim_country/ To exit full screen, press and hold **esc**

Amazon S3

- ▼ Buckets
 - General purpose buckets
 - Directory buckets
 - Table buckets
 - Vector buckets
- ▼ Files
 - File systems [New](#)
- ▼ Access management and security
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dim_country/

Objects (1)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	part-0000-9662c4db-af28-4b4e-854e-dc53be77a001-c000.snappy.parquet	parquet	June 1, 2026, 21:56:02 (UTC-05:00)	756.0 B	Standard

Amazon S3 > Buckets > covid19-analytics-pipeline-kusum > gold/ > dim_date/

Amazon S3

- ▼ Buckets
 - General purpose buckets
 - Directory buckets
 - Table buckets
 - Vector buckets
- ▼ Files
 - File systems [New](#)
- ▼ Access management and security
 - Access Points
 - Access Points for FSx
 - Access Grants
 - IAM Access Analyzer
- ▼ Storage management and insights
 - Storage Lens
 - Batch Operations

Account and organization settings

► AWS Marketplace for S3

dim_date/ [Copy S3 URI](#)

Objects (1) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	part-00000-5cc6c740-3bf2-47df-8657-46d5aee8e471-c000.snappy.parquet	parquet	June 1, 2026, 21:56:04 (UTC-05:00)	1.2 KB	Standard

Amazon S3 > Buckets > covid19-analytics-pipeline-kusum > gold/ > fact_covid_daily/ > year=2024/

Amazon S3

- ▼ Buckets
 - General purpose buckets
 - Directory buckets
 - Table buckets
 - Vector buckets
- ▼ Files
 - File systems [New](#)
- ▼ Access management and security
 - Access Points
 - Access Points for FSx
 - Access Grants
 - IAM Access Analyzer
- ▼ Storage management and insights
 - Storage Lens
 - Batch Operations

Account and organization settings

► AWS Marketplace for S3

year=2024/ [Copy S3 URI](#)

Objects (1) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	month=1/	Folder	-	-	-

STEP 11 – CREATE GOLD CRAWLER

Go to:

Glue → Crawlers → Create crawler

Crawler name:

“covid19_gold_crawler”

Add paths:

s3://covid19-analytics-pipeline-kusum/gold/dim_country/

s3://covid19-analytics-pipeline-kusum/gold/dim_date/

s3://covid19-analytics-pipeline-kusum/gold/fact_covid_daily/

Go to:
Settings → Manage

Query result location:
s3://covid19-analytics-pipeline-kusum/athena-results/

Save settings.

Choose database:
“covid19_db”

Screenshot:

Amazon Athena > Query editor

Settings successfully updated.

Query in Amazon SageMaker Unified Studio
Use your IAM role to analyze and build with your existing AWS resources in a single data and AI development environment. [Set up](#)

Editor | Recent queries | Saved queries | **Query settings** | Workgroup: primary

Query settings

Query settings temporarily override your workgroup configurations. Clear all the query settings to enforce your workgroup configurations.

Query result encryption

Query result location
s3://covid19-analytics-pipeline-kusum/athena-results/

Encrypt query results
-

Expected bucket owner
-

Assign bucket owner full control over query results
Turned off

[Manage](#)

Execution controls

Set limits and control query execution behavior for the workgroup. Controls apply to all queries in the workgroup unless overridden.

Search execution controls

Control Name	Value	Unit
Data scanned limit	-	-
Min DPU per query	-	-
Max DPU per query	-	-

[Edit controls](#)

STEP 13 – RUN ATHENA QUERIES

Run:

```
SELECT country, SUM(new_cases) AS total_new_cases
FROM fact_covid_daily
GROUP BY country;
```

Amazon Athena > Query editor

Athena now supports typeahead code suggestions to speed up SQL query development. Typeahead suggestions are turned on by default. You can change this setting in query editor preferences. [Edit preferences](#)

Data

Data source: AwsDataCatalog

Catalog: None

Database: covid19_db

Tables and views: [Create](#)

Filter tables and views

Tables (9)

- covid_cases
- covid_cases_csv
- covid_testing
- covid_testing_csv
- covid_vaccinations
- covid_vaccinations_csv
- dim_country
- dim_date
- fact_covid_daily

Views (0)

Query 1

```
1 SELECT
2   country,
3   SUM(new_cases) AS total_new_cases
4 FROM fact_covid_daily
5 GROUP BY country
6 ORDER BY total_new_cases DESC;
```

SQL Ln 6, Col 31

[Run again](#) [Explain](#) [Cancel](#) [Clear](#) [Create](#)

[Reuse query results](#) up to 60 minutes ago

Query results **Query stats**

[Completed](#) Time in queue: 60 ms Run time: 587 ms Data scanned: 0.13 KB

[Copy](#) [Download results CSV](#)

Search rows

#	country	total_new_cases
1	India	15000
2	USA	11000
3	Nepal	650

Run:

```
SELECT country, report_date, case_severity
FROM fact_covid_daily
WHERE case_severity = 'High';
```

Amazon Athena > Query editor

[editor](#) [recent queries](#) [save queries](#) [query settings](#) **Workgroup** primary

Athena now supports typeahead code suggestions to speed up SQL query development. Typeahead suggestions are turned on by default. You can change this setting in query editor preferences. [Edit preferences](#)

Data

Data source: AwsDataCatalog

Catalog: None

Database: covid19_db

Tables and views: [Create](#)

Filter tables and views

Tables (9)

- covid_cases
- covid_cases_csv
- covid_testing
- covid_testing_csv
- covid_vaccinations
- covid_vaccinations_csv
- dim_country
- dim_date
- fact_covid_daily

Views (0)

Query 1 **Query 2**

```
1 SELECT
2   country,
3   report_date,
4   case_severity
5 FROM fact_covid_daily
6 WHERE case_severity = 'High';
```

SQL Ln 6, Col 30

[Run again](#) [Explain](#) [Cancel](#) [Clear](#) [Create](#)

[Reuse query results](#) up to 60 minutes ago

Query results **Query stats**

[Completed](#) Time in queue: 99 ms Run time: 578 ms Data scanned: 0.26 KB

[Copy](#) [Download results CSV](#)

Search rows

#	country	report_date	case_severity
1	India	2024-01-02	High
2	India	2024-01-01	High

Run:

```
SELECT country, positivity_rate
FROM fact_covid_daily;
```

Amazon Athena > Query editor

Data

Data source: AwsDataCatalog

Catalog: None

Database: covid19_db

Tables and views: [Create](#)

Filter tables and views

Tables (9)

- covid_cases
- covid_cases_csv
- covid_testing
- covid_testing_csv
- covid_vaccinations
- covid_vaccinations_csv
- dim_country
- dim_date
- fact_covid_daily (Partitioned)

Views (0)

Query 1: `SELECT
country,
report_date,
positivity_rate
FROM fact_covid_daily;`

SQL Ln 5, Col 23

[Run again](#) [Explain](#) [Cancel](#) [Clear](#) [Create](#)

Query results **Query stats**

[Completed](#) Time in queue: 52 ms Run time: 517 ms Data scanned: 0.23 KB

[Copy](#) [Download results CSV](#)

Search rows

#	country	report_date	positivity_rate
1	India	2024-01-02	5.3
2	USA	2024-01-02	6.0
3	India	2024-01-01	4.6
4	Nepal	2024-01-01	6.0
5	USA	2024-01-01	5.0
6	Nepal	2024-01-02	7.0

STEP 14 – CREATE REDSHIFT SERVERLESS

Open Amazon Redshift.

Choose:

Redshift Serverless

Click:

Create workgroup

Use:

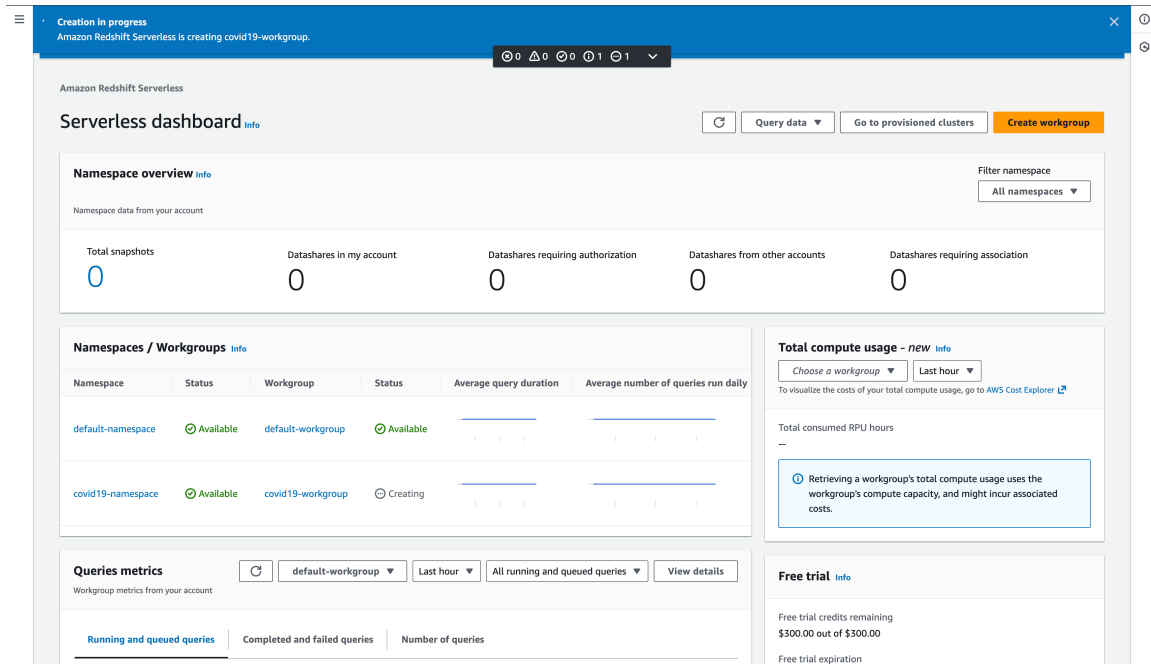
- Namespace: covid19-namespace
- Workgroup: covid19-workgroup
- Database: dev
- Username: admin
- Base capacity: 8 RPUs

Create workgroup.

Wait until:

“Available”

Screenshot:



STEP 15 – OPEN QUERY EDITOR V2

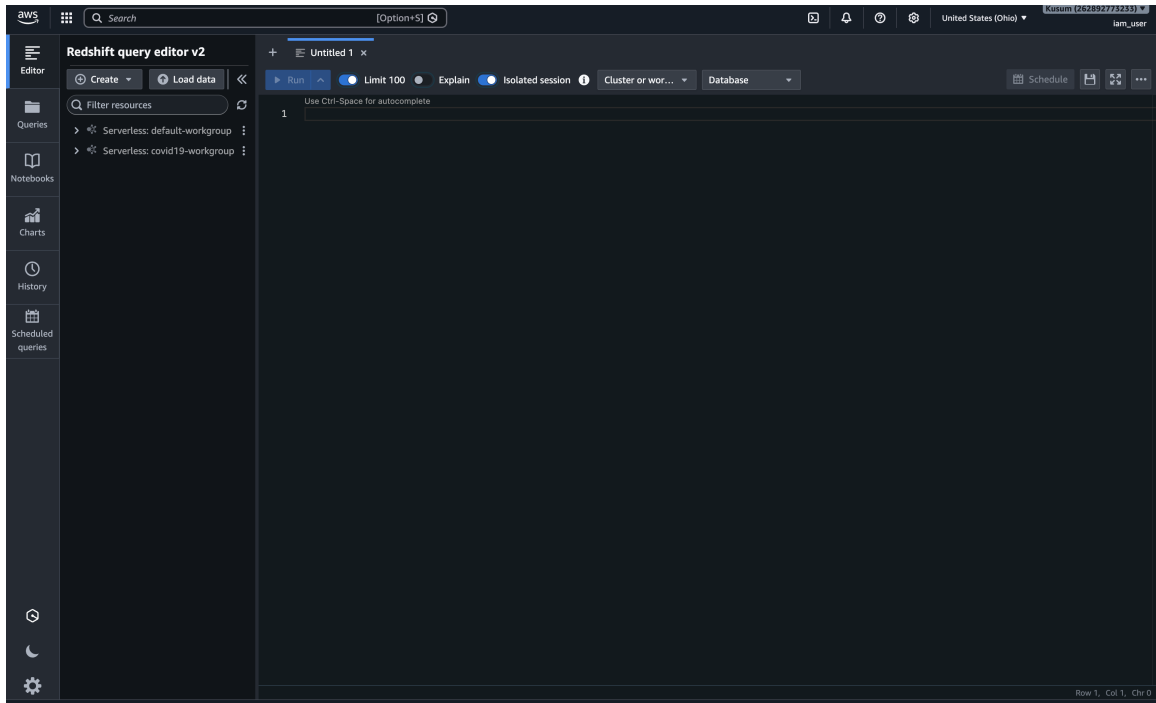
Go to:

Redshift → Query Editor v2

Connect using:

- Workgroup: covid19-workgroup
- Database: dev
- Username: admin
- Password: your password

Screenshot:



=====

STEP 16 – CREATE REDSHIFT TABLES

Create:

- dim_country
- dim_date
- fact_covid_daily

Screenshot:

Redshift query editor v2

Editor

Filter resources

Serverless: default-workgroup

Serverless: covid19-workgr...

native databases (2)

dev

public

Tables 3

dim...

di...

fa...

Views 0

f/ Funci... 0

Stored... 0

dim_country

Field	Type	NL
A country	character varying(100)	NULL
country_id	bigint	NULL

Summary

Returned rows: 0

Elapsed time: 122ms

Result set query:

```
CREATE TABLE dim_date (  
  report_date DATE,  
  year INT,  
  month INT,  
  date_id BIGINT  
);  
  
CREATE TABLE fact_covid_daily (  
  country_id BIGINT,  
  date_id BIGINT,  
  country VARCHAR(100),  
  report_date DATE,  
  total_cases BIGINT,  
  new_cases BIGINT,  
  total_deaths BIGINT,  
  new_deaths BIGINT,  
  total_vaccinations BIGINT,  
  people_vaccinated BIGINT,  
  people_fully_vaccinated BIGINT  
);
```

Row 31, Col 3, Chr 636

Result 1 Result 2 Result 3

Export Chart

Redshift query editor v2

Editor

Filter resources

Serverless: default-workgroup

Serverless: covid19-workgr...

native databases (2)

dev

public

Tables 3

dim...

di...

fa...

Views 0

f/ Funci... 0

Stored... 0

dim_country

Field	Type	NL
A country	character varying(100)	NULL
country_id	bigint	NULL

Summary

Returned rows: 0

Elapsed time: 139ms

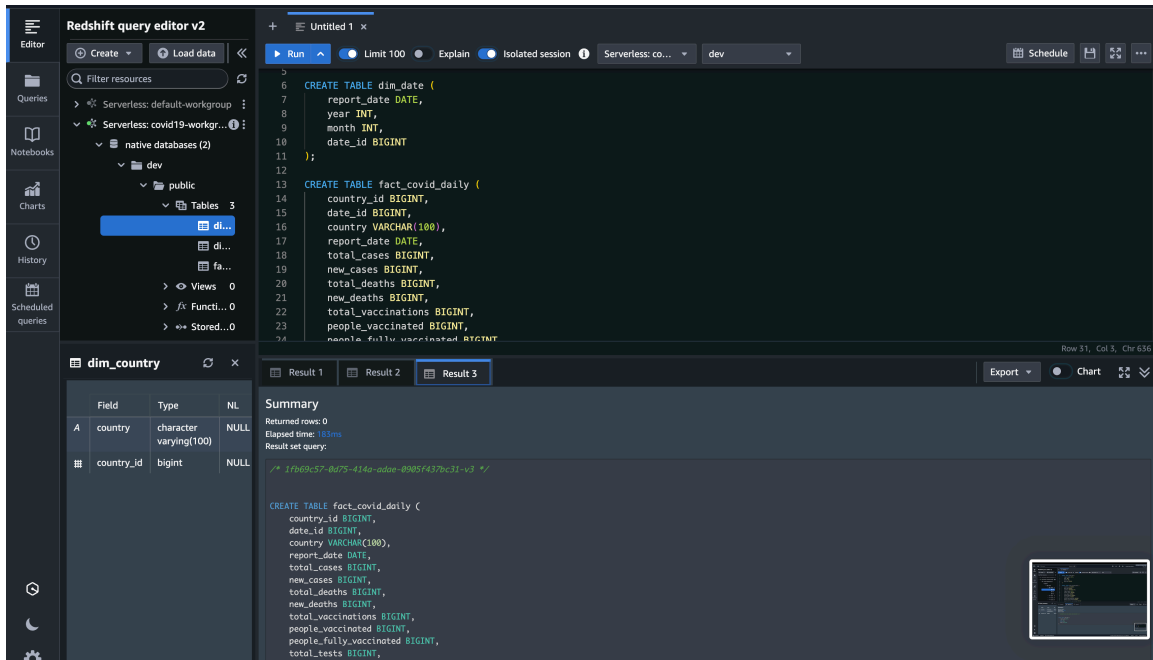
Result set query:

```
CREATE TABLE dim_date (  
  report_date DATE,  
  year INT,  
  month INT,  
  date_id BIGINT  
);  
  
CREATE TABLE fact_covid_daily (  
  country_id BIGINT,  
  date_id BIGINT,  
  country VARCHAR(100),  
  report_date DATE,  
  total_cases BIGINT,  
  new_cases BIGINT,  
  total_deaths BIGINT,  
  new_deaths BIGINT,  
  total_vaccinations BIGINT,  
  people_vaccinated BIGINT,  
  people_fully_vaccinated BIGINT  
);
```

Row 31, Col 3, Chr 636

Result 1 Result 2 Result 3

Export Chart



STEP 17 – LOAD GOLD DATA INTO REDSHIFT

Get IAM Role ARN from:

IAM → Roles

Run COPY commands to load data from:

- s3://covid19-analytics-pipeline-kusum/gold/

Screenshot:

IAM

> Roles

Identity and Access Management (IAM)

Q Search IAM

Dashboard

▼ Access Management

Roles

Policies

IAM users

IAM user groups

Identity providers

Account settings

Root access management

Temporary delegation requests

▼ Access reports

Access Analyzer

Resource analysis

Unused access

Analyzer settings

Credential report

Organization activity

Service control policies

Resource control policies

Related consoles

IAM Identity Center

AWS Organizations

Role redshift_role created.

View role

Roles (7)

info

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Q Search

☐

Role name

▲

Trusted entities

Last activity

▼

☐

[AWSServiceRoleForResourceExplorer](#)

AWS Service: resource-explorer-2 (S

16 minutes ago

☐

[AWSServiceRoleForSupport](#)

AWS Service: support (Service-Linker

-

☐

[AWSServiceRoleForTrustedAdvisor](#)

AWS Service: trustedadvisor (Service

-

☐

[GlueCovidDataPlatformRole](#)

AWS Service: glue

9 minutes ago

☐

[myLambda](#)

AWS Service: lambda

-

☐

[redshift_role](#)

AWS Service: redshift

-

☐

[s3-roles](#)

AWS Service: ec2

21 days ago

Roles Anywhere

info

Authenticate your non AWS workloads and securely provide access to AWS services.

0.0

0.0

0.0

Access AWS from your non AWS workloads

Operate your non AWS workloads using the same authentication and authorization strategy that you use within AWS.

X.509 Standard

Use your own existing PKI infrastructure or use [AWS Certificate Manager Private Certificate Authority](#) to authenticate identities.

Temporary credentials

Use temporary credentials with ease and benefit from the enhanced security they provide.

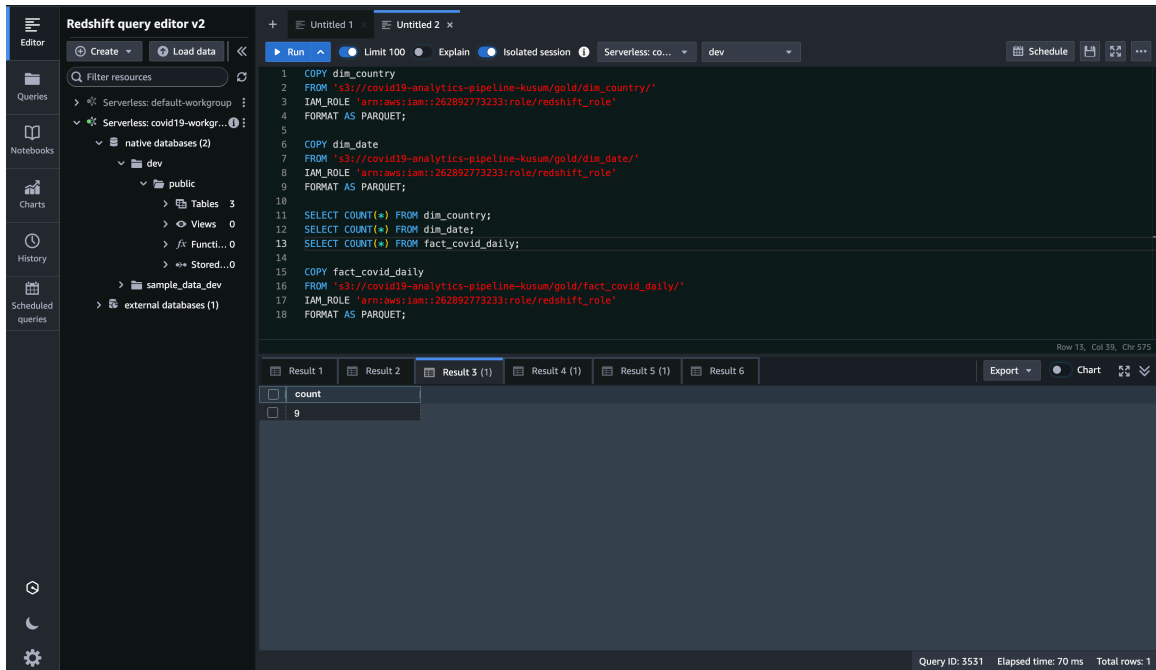
=====

Step 18 — Verify Redshift Data Load

I checked the number of records loaded into Redshift tables.

```
SELECT COUNT(*) FROM dim_country;
SELECT COUNT(*) FROM dim_date;
SELECT COUNT(*) FROM fact_covid_daily;
```

Screenshot to attach:



STEP 19 – RUN BI QUERIES

Run reporting queries:

- Country summary report
- Severity report
- Vaccination analytics report

Screenshots:

Redshift query editor v2

Editor

Create Load data

Filter resources

Serverless: default-workgroup

Serverless: covid19-work...

native databases (2)

external databases (1)

Queries

Notebooks

Charts

History

Scheduled queries

Run Limit 100 Explain Isolated session Serverless: co... dev

Schedule

1 SELECT

2 country,

3 SUM(new_cases) AS total_new_cases,

4 SUM(new_deaths) AS total_new_deaths

5 FROM fact_covid_daily

6 GROUP BY country

7 ORDER BY total_new_cases DESC;

8

9 SELECT

10 case_severity,

11 COUNT(*) AS record_count

12 FROM fact_covid_daily

13 GROUP BY case_severity;

Result Result

Export Chart

case_severity	record_count
---------------	--------------

No Rows To Show

Row 15, Col 24, Chr 270

Redshift query editor v2

Editor

Create Load data

Filter resources

Serverless: default-workgroup

Serverless: covid19-work...

native databases (2)

external databases (1)

Queries

Notebooks

Charts

History

Scheduled queries

Run Limit 100 Explain Isolated session Serverless: co... dev

Schedule

3 SUM(new_cases) AS total_new_cases,

4 SUM(new_deaths) AS total_new_deaths

5 FROM fact_covid_daily

6 GROUP BY country

7 ORDER BY total_new_cases DESC;

8

9 SELECT

10 case_severity,

11 COUNT(*) AS record_count

12 FROM fact_covid_daily

13 GROUP BY case_severity;

14

15 SELECT

16 country,

17 report_date,

18 people_fully_vaccinated,

19 positivity_rate

20 FROM fact_covid_daily

21 ORDER BY country;

Result 1 Result 2 Result 3

Export Chart

Summary

Returned rows: 0

Elapsed time: 5ms

Result set query:

```
/* IN202308-034d-43f4-9fc9-uc8du20945d7-v3 */
SELECT
  country,
  SUM(new_cases) AS total_new_cases,
  SUM(new_deaths) AS total_new_deaths
FROM fact_covid_daily
GROUP BY country
ORDER BY total_new_cases DESC limit 100;
```

Redshift query editor v2

Editor

Filter resources

Serverless: covid19-work...

native databases (2)

external databases (1)

Queries

Notebooks

Charts

History

Scheduled queries

Run

Limit 100

Explain

Isolated session

Serverless: co... dev

Schedule

Export

Chart

Summary

Returned rows: 0

Elapsed time: 45ms

Result set query:

```
/* F43cc635-86cf-40fa-9954-85ee47928f4d-v3 */

SELECT
  case_severity,
  COUNT(*) AS record_count
FROM fact_covid_daily
GROUP BY case_severity;

SELECT
  country,
  report_date,
  people_fully_vaccinated,
  positivity_rate
FROM fact_covid_daily
ORDER BY country;
```

Row 21, Col 18, Chr 397

Amazon S3 > Buckets > aws-glue-assets-262892773233-us-east-2 > scripts/

scripts/

Copy S3 URI

Objects Properties

Objects (2)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	covid19_silver_g old_etl_job.py	py	June 1, 2026, 21:50:26 (UTC- 05:00)	4.8 KB	Standard
☐	healthcare_etl_jo b.py	py	May 24, 2026, 00:06:08 (UTC- 05:00)	2.5 KB	Standard

Amazon S3 > Buckets > aws-glue-assets-262892773233-us-east-2 > sparkHistoryLogs/

sparkHistoryLogs/

Copy S3 URI

Objects Properties

Objects (36)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	appstatus_spark- application- 1779598755202	-	May 23, 2026, 23:59:24 (UTC- 05:00)	0 B	Standard
☐	appstatus_spark- application- 1779598874216	-	May 24, 2026, 00:01:25 (UTC- 05:00)	0 B	Standard
☐	appstatus_spark- application- 1779599197373	-	May 24, 2026, 00:07:15 (UTC- 05:00)	0 B	Standard
☐	appstatus_spark- application- 1779599197373. inprogress	inprogress	May 24, 2026, 00:07:05 (UTC- 05:00)	0 B	Standard
☐	appstatus_spark- application- 1780366995005	-	June 1, 2026, 21:23:37 (UTC- 05:00)	0 B	Standard
☐	appstatus_spark- application- 1780368636420	-	June 1, 2026, 21:51:47 (UTC- 05:00)	0 B	Standard
☐	appstatus_spark- application- 1780368636420.	inprogress	June 1, 2026, 21:51:33 (UTC- 05:00)	0 B	Standard

FINAL SCREENSHOT CHECKLIST

- S3 bucket
- Folder structure
- Uploaded CSV files
- IAM role
- Glue database
- Glue crawler
- Glue tables
- ETL script
- ETL success
- Silver output
- Gold output
- Athena queries
- Redshift workgroup
- Redshift tables
- COPY command
- BI reports

FINAL CONCLUSION:

This project successfully implemented an end-to-end AWS COVID-19 analytics platform using Amazon S3, AWS Glue, Athena, and Amazon Redshift. Raw datasets were transformed into Silver-layer parquet datasets and Gold-layer dimensional models for scalable analytics and SQL-based reporting.